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November 9, 2026

The Editor-in-Chief
Transportation Science

Dear Editor,

I submit the manuscript *SORTIE: Fair Vehicle Routing When Route Durations Are Random* for consideration as an article in *Transportation Science*.

Recent exact work on fair vehicle routing combines ordered fairness measures (range, Gini mean difference, mean absolute deviation) with a bilevel requirement that every route be a shortest tour of its customers. That work assumes deterministic travel times. Drivers experience realised durations, and the fairness they perceive is the fairness of the realised day. The manuscript studies this setting and makes four contributions.

1. It separates ex-ante from ex-post fairness and bounds the gap between them, the *fairness illusion*, in terms of the routes' quadratic lengths. It also proves that multiplicative common shocks, such as weather or a fleet-wide congestion factor, leave expected ordered fairness unchanged.
2. It gives an exact, sampling-free formula for the expected value of any ordered measure of independent route durations, including the negative weights of dispersion measures. The formula is a Poisson–binomial order-statistic integral. Its derivatives, the *rank prices*, show which routes a fairness objective would like to lengthen.
3. It shows that, under uncertainty, even the min–max objective can prefer a tour that is not the shortest. Route consistency must therefore be imposed explicitly. It is defined through each driver's own mean–standard-deviation criterion. Such tours are supported Pareto tours and can be computed by parametric Held–Karp.
4. It proposes SORTIE, a bilevel heuristic that screens moves with the rank prices. SORTIE is evaluated on instances derived from the CVRPLIB benchmark used by the deterministic literature, against deterministic and sample-average baselines, under both homogeneous and congestion-driven heterogeneous noise.

Code, instances, solved plans and raw results will be made publicly available.

The manuscript is original, has not been published, and is not under consideration elsewhere. I

have no conflicts of interest to declare.

Yours sincerely,

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